

Enterprise VLAN & Trunking Lab

Phase 5 Report – Network Verification & Testing

Project Information

Project Name: Enterprise VLAN & Trunking Lab

Phase: Phase 5 – Network Verification & Testing

Simulation Software: Cisco Packet Tracer

Devices Verified: SW1, SW2, Eight End Devices

Status: Completed

1. Introduction

After completing the configuration of VLANs, access ports, and the IEEE 802.1Q trunk link, the final implementation step was to verify that the enterprise network operated according to the design specifications.

The purpose of this phase was to validate every major component of the Layer 2 infrastructure, including VLAN membership, trunk operation, interface status, MAC address learning, and end-to-end connectivity. Verification ensures that the deployed configuration is functioning correctly before the network is considered ready for production.

2. Objectives

The objectives of this phase were to:

- Verify the VLAN database on both switches.
 - Validate access port assignments.
 - Confirm IEEE 802.1Q trunk operation.
 - Verify interface operational status.
 - Confirm MAC address learning.
 - Test end-device connectivity.
 - Validate VLAN isolation.
 - Confirm that the running configuration matched the intended network design.
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3. Verification Methodology

A structured verification process was followed using Cisco IOS diagnostic commands and connectivity tests.

The verification focused on:

- VLAN configuration
- Trunk configuration
- Switchport configuration
- Interface status
- MAC address tables
- Connectivity testing
- Running configuration review

Each verification step was documented to ensure the network complied with enterprise deployment standards.

4. VLAN Verification

The VLAN database was verified using:

``text id="vlanverify" show vlan brief

The results confirmed that:

- VLAN 10 (MANAGEMENT) was present.
- VLAN 20 (HR) was present.
- VLAN 30 (SALES) was present.
- VLAN 40 (IT) was present.
- VLAN 99 (NATIVE) was present.
- All access interfaces belonged to their designated VLANs.

SW1

VLAN	Interfaces
10	Fa0/1, Fa0/2
20	Fa0/3, Fa0/4

SW2

VLAN	Interfaces
30	Fa0/1, Fa0/2

```
| 40 | Fa0/3, Fa0/4 |
```

The VLAN database matched the planned enterprise network design.

5. Trunk Verification

The inter-switch trunk was verified using:

```
```text id="trunkverify"
show interfaces trunk
```

Verification confirmed:

- GigabitEthernet0/1 operating in trunk mode.
- IEEE 802.1Q encapsulation active.
- Native VLAN configured as VLAN 99.
- Allowed VLANs:
  - VLAN 10
  - VLAN 20
  - VLAN 30
  - VLAN 40
  - VLAN 99

The trunk link was fully operational.

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## 6. Interface Status Verification

Interface status was verified using:

```
```text id="statusverify" show interfaces status
```

Results confirmed:

- All FastEthernet interfaces connected to end devices were operational.
- GigabitEthernet0/1 was operational.
- No interfaces were administratively down.
- No interfaces were in an error-disabled state.

The physical Layer 1 and Layer 2 connections were functioning correctly.

7. Switchport Verification

The trunk interface configuration was verified using:

```
```text id="switchportverify"
show interfaces gigabitEthernet0/1 switchport
```

The output confirmed:

- Administrative Mode: Trunk
- Operational Mode: Trunk
- Native VLAN: 99
- Allowed VLANs: 10,20,30,40,99

The switchport configuration matched the intended design.

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## 8. MAC Address Table Verification

The MAC address table was examined using:

```
```text id="macverify" show mac address-table
```

The switches successfully learned the MAC addresses of connected hosts.

Dynamic MAC address learning confirmed that:

- End devices were actively communicating with their switches.
- Switching tables were being populated correctly.
- Layer 2 forwarding was operating as expected.

9. Connectivity Testing

ICMP echo requests (ping) were used to validate connectivity between hosts.

Successful Tests

Communication between hosts belonging to the same VLAN was successful.

Examples:

Source	Destination	Result
MGMT-PC1	MGMT-PC2	Success
HR-PC1	HR-PC2	Success
SALES-PC1	SALES-PC2	Success
IT-PC1	IT-PC2	Success

These results confirmed proper Layer 2 forwarding within each VLAN.

Failed Tests

Communication between different VLANs failed.

Examples:

Source	Destination	Result
MGMT-PC1	HR-PC1	Failed
MGMT-PC1	SALES-PC1	Failed
HR-PC1	IT-PC1	Failed
SALES-PC1	IT-PC1	Failed

These failures were expected.

Each VLAN represents an independent Layer 2 broadcast domain, and no Layer 3 routing device had been deployed in the network. Consequently, hosts belonging to different VLANs could not exchange IP traffic.

The failed ping tests therefore confirmed that VLAN isolation was functioning correctly.

10. Running Configuration Verification

The complete switch configuration was reviewed using:

```
```text id="runningverify"
show running-config
```

The review confirmed:

- Correct hostnames.
- VLAN database consistency.

- Proper access port assignments.
- Trunk configuration.
- Native VLAN configuration.
- Allowed VLAN list.
- Password configuration.
- Security banner.
- Interface descriptions.

The running configuration accurately reflected the planned enterprise topology.

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## 11. Network Validation Summary

The following components were successfully validated:

Component	Status
VLAN Configuration	Verified
Access Ports	Verified
IEEE 802.1Q Trunk	Verified
Native VLAN	Verified
Allowed VLAN List	Verified
Interface Status	Verified
MAC Address Learning	Verified
Intra-VLAN Connectivity	Verified
Inter-VLAN Isolation	Verified
Running Configuration	Verified

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## 12. Technical Concepts Reinforced

This phase strengthened practical knowledge of:

- Enterprise network verification
- VLAN validation
- IEEE 802.1Q trunk verification
- Cisco IOS diagnostic commands
- MAC address learning
- Layer 2 forwarding

- Broadcast domain isolation
- Connectivity testing
- Network documentation

These skills are essential for network engineers responsible for deploying and validating enterprise campus networks.

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## 13. Best Practices Applied

The following operational best practices were followed:

- Verified every configuration after deployment.
  - Used Cisco diagnostic commands rather than assumptions.
  - Performed structured connectivity testing.
  - Validated both successful and unsuccessful communication scenarios.
  - Confirmed MAC address learning before concluding the deployment.
  - Saved configurations after successful verification.
  - Documented verification results for future reference.
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## 14. Lessons Learned

Several important networking principles were demonstrated during this phase:

- Configuration must always be followed by verification.
  - Successful communication within a VLAN confirms correct Layer 2 switching.
  - Failed communication between different VLANs confirms proper VLAN isolation.
  - MAC address tables provide valuable insight into switch forwarding behavior.
  - Trunk links can be operational even when no user traffic is currently traversing multiple switches.
  - Verification commands are essential tools for diagnosing and validating enterprise networks.
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## 15. Preparation for Phase 6

The network has now been successfully deployed and validated.

The next phase will focus on troubleshooting techniques, including:

- Diagnosing VLAN misconfigurations.
- Identifying trunk configuration errors.
- Detecting native VLAN mismatches.
- Resolving connectivity failures.
- Applying systematic troubleshooting methodologies used by enterprise network engineers.

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## 16. Conclusion

Phase 5 successfully validated the complete Layer 2 enterprise network deployment. All VLANs, access ports, trunk links, interface states, and MAC address tables were verified using Cisco IOS diagnostic commands. Connectivity tests confirmed that communication within the same VLAN functioned correctly while communication between different VLANs remained isolated, as expected in a Layer 2 environment without inter-VLAN routing.

The successful completion of this phase demonstrates that the enterprise switching infrastructure has been correctly implemented and verified according to industry best practices, providing a reliable foundation for future enhancements such as inter-VLAN routing and advanced network services.